

1 Propositions, connectives and conditional statements

Source: *Unit_2_Logic.pdf*, sections 1–2.

Scope of this rewrite. This edition uses only the procedures in the two attached files, *Unit_2_Logic.pdf* and *Unit_3_Combinatorics.pdf*. It follows their examples and notation. Formulas precede solutions; evident typing/index errors are corrected without replacing the method. Induction is listed but not developed in Unit 2; see page 5.

A **proposition** is a declarative sentence with exactly one truth value, true (T) or false (F). A question, command, or open statement such as $x > 3$ with unspecified x is not a proposition. Letters p, q denote whole propositions.

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

$\neg$ means NOT; $\wedge$ means AND; $\vee$ means inclusive OR; $\rightarrow$ means “if...then”; $\leftrightarrow$ means “if and only if”. The conditional is false only when p is true and q is false. The biconditional is true when the values agree.

English translation. “But” means AND. “ p only if q ”, “ p is sufficient for q ”, and “ q is necessary for p ” all mean $p \rightarrow q$. “Either...or” is inclusive unless exclusivity is specified.

Negate a comparison

Question. Negate “Rohan is richer than Shankar.” **Formula / rule first.** $\neg(a > b) \equiv a \leq b$, where a, b are the compared quantities. Let a be Rohan’s wealth and b Shankar’s. The negation is **Rohan is not richer than Shankar**; he may have equal or less wealth. Similarly, “Sangeeta sleeps longer than Sujata” becomes “Sangeeta does not sleep longer than Sujata.”

Remove a conditional; form its related statements

Question. Use “If it is cold, he wears a hat.” Let p mean “it is cold” and q mean “he wears a hat”. **Formula / rule first.** $p \rightarrow q \equiv \neg p \vee q$; converse $q \rightarrow p$; inverse $\neg p \rightarrow \neg q$; contrapositive $\neg q \rightarrow \neg p$.

Form	Result
Without conditional	It is not cold or he wears a hat.
Converse	If he wears a hat, then it is cold.
Inverse	If it is not cold, then he does not wear a hat.
Contrapositive	If he does not wear a hat, then it is not cold.

The original and contrapositive are equivalent. The converse and inverse are equivalent to each other, but generally not to the original.

“Only if”: Marc’s test

Question. Find the contrapositive of “Only if Marc studies will he pass the test.” **Formula / rule first.** “ p only if q ” is $p \rightarrow q$; its contrapositive is $\neg q \rightarrow \neg p$. Let p mean Marc passes and q mean Marc studies. The statement is $p \rightarrow q$. Therefore: **if Marc does not study, he will not pass the test.**

Same source pattern: “If Erik is a poet, he is poor” has contrapositive “If Erik is not poor, he is not a poet.” Interchange and negate both parts.

2 Laws, classifications and normal forms

Source: Unit 2, sections 3–4 and 7.

Logical equivalence means matching truth values for every assignment. $P \equiv Q$ denotes equivalence; capital letters here may represent compound formulas.

Law	Equivalent forms
Identity	$p \wedge T \equiv p; p \vee F \equiv p$
Domination	$p \vee T \equiv T; p \wedge F \equiv F$
Idempotent	$p \vee p \equiv p; p \wedge p \equiv p$
Double negation	$\neg\neg p \equiv p$
Commutative	$p \vee q \equiv q \vee p; p \wedge q \equiv q \wedge p$
Associative	$(p \vee q) \vee r \equiv p \vee (q \vee r)$; similarly for $\wedge$
Distributive	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$ $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
De Morgan	$\neg(p \wedge q) \equiv \neg p \vee \neg q; \neg(p \vee q) \equiv \neg p \wedge \neg q$
Implication	$p \rightarrow q \equiv \neg p \vee q$
Biconditional	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
Absorption	$p \vee (p \wedge q) \equiv p; p \wedge (p \vee q) \equiv p$

Verify identities and classify formulas by truth table

Formula / rule first. Equivalent columns must agree in every row. A tautology is always T; a contradiction is always F; a contingency has both values. Satisfiable means T in at least one row.

p	$p \wedge T$	$p \vee F$	$p \wedge F$	$p \vee T$	$p \vee \neg p$	$p \wedge \neg p$
T	T	T	F	T	T	F
F	F	F	F	T	T	F

The first two expression columns equal p ; the next two are always F and always T. Thus $p \vee \neg p$ is a **tautology**, and $p \wedge \neg p$ is a **contradiction**. A tautology is satisfiable; a contradiction is not.

DNF and CNF: the source's conversion procedure

A **literal** is a proposition letter or its negation. **DNF** is an OR of AND-terms, for example $(p \wedge q) \vee (\neg p \wedge r)$. **CNF** is an AND of OR-clauses, for example $(p \vee q) \wedge (\neg p \vee r)$.

Formula / rule first. Use the biconditional, implication, De Morgan, double-negation and distributive laws in the table.

1. Eliminate $\leftrightarrow$: $p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$.
2. Eliminate $\rightarrow$: $p \rightarrow q \equiv \neg p \vee q$.
3. Push negations inward using De Morgan and double negation.
4. For **DNF**, distribute AND over OR until OR is the outside operation.
5. For **CNF**, distribute OR over AND until AND is the outside operation.
6. Simplify using identity, idempotent and absorption laws where applicable.

Source limitation. Unit 2 explicitly says the original expressions to convert are missing. This edition preserves its definitions and conversion procedure; it does not invent those missing classwork questions or introduce principal/canonical-form methods.

3 Validity and the source's rules of inference

Source: Unit 2, section 5.

An argument is **valid** if its conclusion is true whenever all premises are true. Validity concerns the form of the reasoning.

Rule	Premises $\implies$ conclusion
Modus Ponens (MP)	$p, p \rightarrow q \implies q$
Modus Tollens (MT)	$\neg q, p \rightarrow q \implies \neg p$
Hypothetical Syllogism (HS)	$p \rightarrow q, q \rightarrow r \implies p \rightarrow r$
Disjunctive Syllogism	$p \vee q, \neg p \implies q$
Addition	$p \implies p \vee q$
Simplification	$p \wedge q \implies p$
Conjunction	$p, q \implies p \wedge q$
Proof by cases	$p \vee q, p \rightarrow r, q \rightarrow r \implies r$

Classwork: study, failure and basketball

Question. If I study, I will not fail mathematics. If I do not play basketball, I study. I failed mathematics. Therefore I played basketball. Test validity. **Formula / rule first.** MT: $p \rightarrow q, \neg q \implies \neg p$; double negation: $\neg\neg p \equiv p$. Let S mean study, F mean fail mathematics, and B mean play basketball.

Step	Statement	Reason
1	$S \rightarrow \neg F$	Premise
2	$\neg B \rightarrow S$	Premise
3	F	Premise
4	$\neg S$	MT on 1 and 3, since $F \equiv \neg\neg F$
5	$\neg\neg B$	MT on 2 and 4
6	B	Double negation

Conclusion: valid. In the source's cinema variant, the premises are $S \rightarrow P, \neg C \rightarrow S, \neg P$. The same MT rule gives $\neg S$, then $\neg\neg C$, hence C : I went to the cinema.

Classwork: tigers, mammals and scales

Formula / rule first. HS: $p \rightarrow q, q \rightarrow r \implies p \rightarrow r$; apply the universal statements to an arbitrary object and then generalize. Let $T(x)$ mean tiger, $M(x)$ mammal, and $S(x)$ having scales. The premises are $\forall x(T(x) \rightarrow M(x))$ and $\forall x(M(x) \rightarrow \neg S(x))$. For arbitrary x , $T(x) \rightarrow M(x)$ and $M(x) \rightarrow \neg S(x)$. HS gives $T(x) \rightarrow \neg S(x)$. Generalizing gives $\boxed{\forall x(T(x) \rightarrow \neg S(x))}$. Thus no tigers have scales, and the argument is valid.

Classwork: two invalid forms

Formula / rule first. Neither $p \rightarrow q, q \implies p$ nor $p \rightarrow q, \neg p \implies \neg q$ is a valid rule.

- Every problem / learning mathematics.** Put p = do every problem and q = learn discrete mathematics. The stated form is $p \rightarrow q, q \therefore p$: **affirming the consequent**, invalid.
- Equal sides / equal opposite angles.** Put p = the two sides are equal and q = their opposite angles are equal. The stated form is $p \rightarrow q, \neg p \therefore \neg q$: **denying the antecedent**, invalid as an inference from these premises.

Translate first, identify the inference rule second, and give the conclusion last. Do not reverse an implication merely because its conclusion is given.

4 Predicates, quantifiers and nested translations

Source: Unit 2, section 6.

A **predicate** contains variables; it becomes a proposition when its variables are assigned values or quantified. $\forall x$ means “for every x ”; $\exists x$ means “there exists at least one x ”. Always state the domain.

Translate the source’s three statements

Formula / rule first. “Every P is Q ” becomes $\forall x(P(x) \rightarrow Q(x))$. Existence of a suitable value is expressed using $\exists$.

1. **The sum of two positive integers is positive.** Domain: $\mathbb{Z}$. Choose arbitrary integers x, y ; require the conclusion when both are positive:

$$\boxed{\forall x \forall y [((x > 0) \wedge (y > 0)) \rightarrow x + y > 0].}$$

2. **Every nonzero real number has a multiplicative inverse.** Domain: $\mathbb{R}$. For each $x \neq 0$, a value y must satisfy $xy = 1$:

$$\boxed{\forall x [x \neq 0 \rightarrow \exists y (xy = 1)].}$$

3. **Every student in this class has studied calculus.** Let $S(x)$ mean student in this class, $C(x)$ mean has studied calculus, and take people as the domain:

$$\boxed{\forall x [S(x) \rightarrow C(x)].}$$

Nested quantifiers: computers and friendship

Question. Interpret $\forall x(C(x) \vee \exists y(C(y) \wedge F(x, y)))$, where the domain is students, $C(x)$ means has a computer, and $F(x, y)$ means x, y are friends. **Formula / rule first.** Read quantifiers from outside inward; $\vee$ means OR and $\wedge$ requires both conditions.

1. $\forall x$: consider each student x .
2. $C(x) \vee \dots$: either that student has a computer, or the second condition holds.
3. $\exists y(C(y) \wedge F(x, y))$: some student y has a computer and is a friend of x .

Answer: every student either has a computer or has a friend who has a computer.

Nested quantifiers: a friendship structure

Question. Interpret $\exists x \forall y \forall z [(F(x, y) \wedge F(x, z) \wedge y \neq z) \rightarrow \neg F(y, z)]$. **Formula / rule first.** $\exists x$ selects a particular student; $\forall y \forall z$ covers every pair; implication restricts the conclusion to pairs satisfying all three conditions. There exists a student x such that whenever two distinct students y, z are both friends of x , they are not friends of each other. The repeated x refers to the same selected student throughout.

Negate “Every student has a laptop”

Formula / rule first. $\neg \forall x P(x) \equiv \exists x \neg P(x)$; $\neg \exists x P(x) \equiv \forall x \neg P(x)$; $\neg(p \rightarrow q) \equiv p \wedge \neg q$ (by implication and De Morgan). Let $S(x)$ mean student and $L(x)$ mean has a laptop. Then

$$\neg \forall x (S(x) \rightarrow L(x)) \equiv \exists x \neg (S(x) \rightarrow L(x)) \equiv \boxed{\exists x (S(x) \wedge \neg L(x))}.$$

Answer: there exists a student who does not have a laptop.

Keep the source’s order of quantifiers and the brackets that determine their scope. Translate one logical layer at a time.

5 Proof by contraposition and contradiction

Source: Unit 2, section 8; induction coverage checked against all 10 source pages.

Definitions. An even integer is $2k$ and an odd integer is $2k + 1$, with $k \in \mathbb{Z}$. A rational number is a/b for integers a, b with $b \neq 0$; in lowest terms they have no common factor greater than 1.

Contraposition: if n^2 is odd, then n is odd

Formula / rule first. $p \rightarrow q \equiv \neg q \rightarrow \neg p$; even $n = 2k$.

1. The contrapositive is: if n is even, then n^2 is even.
2. Assume $n = 2k$ for an integer k .
3. Compute $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$.
4. Since $2k^2 \in \mathbb{Z}$, the square is even. Hence the contrapositive, and therefore the original statement, is true.

Contraposition: if $3n + 2$ is odd, then n is odd

Formula / rule first. Again prove $\neg q \rightarrow \neg p$, with even $n = 2k$. Assume n is even. Then

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1).$$

Since $3k + 1$ is an integer, $3n + 2$ is even. This proves the contrapositive and the required implication.

Contradiction: $\sqrt{2}$ is irrational

Formula / rule first. Assume the negation of the claim and derive an impossibility. A rational number has a lowest-terms representation a/b ; an even integer square has an even integer root.

1. Assume $\sqrt{2} = a/b$ in lowest terms, with $a, b \in \mathbb{Z}$, $b \neq 0$.
2. Square: $a^2 = 2b^2$. Hence a is even; write $a = 2k$.
3. Substitute: $4k^2 = 2b^2$, so $b^2 = 2k^2$. Hence b is even.
4. Both a, b are divisible by 2, contradicting lowest terms. Therefore $\boxed{\sqrt{2} \text{ is irrational}}.$

Contradiction: $\sqrt{5}$ is irrational

Formula / rule first. Use a lowest-terms rational representation. The divisibility fact used in the source is: for prime p , $p \mid a^2 \Rightarrow p \mid a$.

1. Assume $\sqrt{5} = a/b$ in lowest terms. Squaring gives $a^2 = 5b^2$.
2. Thus $5 \mid a$, so put $a = 5k$.
3. Substitute: $25k^2 = 5b^2$, hence $b^2 = 5k^2$, so $5 \mid b$.
4. Both numerator and denominator are divisible by 5, a contradiction. Thus $\boxed{\sqrt{5} \text{ is irrational}}.$

Mathematical induction: gap in the attachment. It appears in Unit 2's title, coverage and learning outcomes, but the file contains no induction section, procedure or worked example. No induction method has been supplied from another source in this rewrite. This remains a gap to resolve from your class material.

6 Answer extraction: match facts, unify, report the value

Source: Unit 2, section 9.

The attached notes define **answer extraction** as finding the value of a query variable by matching the query with knowledge-base facts and applying inference if a rule is involved. **Unification** finds substitutions that make the predicate and its arguments match.

1. Represent the facts and any rules using predicates.
2. Write the query with the unknown variable.
3. Compare the query with the facts.
4. Match the predicate name and all fixed arguments.
5. Substitute the matching value for the variable; apply inference if needed.
6. Report that value as the answer.

Classwork: who knows Python?

Facts. Rahul is a Computer Engineering student; Rahul knows Python; Priya knows Java. **Formula / rule first.** A query $P(x, c)$ matches $P(a, c)$ under the substitution $x = a$; both the predicate name and fixed argument c must agree.

1. Write Student(Rahul), Knows(Rahul, Python), and Knows(Priya, Java).
2. Query: Knows(x , Python)?
3. The matching fact is Knows(Rahul, Python).
4. Unify $x = \text{Rahul}$. **Answer: Rahul knows Python.**

The student fact has the wrong predicate; the Java fact has the wrong language. Neither answers this query.

Classwork: who knows Java?

Facts. Anjali knows C; Kiran knows Python; Meera knows Java; Rohit knows C++. **Formula / rule first.** Match Knows(x , Java) with Knows(a , Java) and substitute $x = a$. Write each fact as Knows(person, language). Only Knows(Meera, Java) matches. Therefore $x = \text{Meera}$.

Practice: who owns a Tablet?

Facts. Ravi owns a Laptop; Sneha owns a Tablet; Arjun owns a Smartphone; Neha owns a Smartwatch. **Formula / rule first.** Owns(x , Tablet) matches Owns(a , Tablet) with $x = a$. The matching fact is Owns(Sneha, Tablet). Unification gives $x = \text{Sneha}$.

Practice: who lives in Chennai?

Facts. Aman lives in Delhi; Priya in Mumbai; Karthik in Chennai; Neha in Pune. **Formula / rule first.** Lives(x , Chennai) matches Lives(a , Chennai) with $x = a$. Query Lives(x , Chennai); match Lives(Karthik, Chennai); conclude $x = \text{Karthik}$.

Practice: who manufactures Camera?

Facts. Sony manufactures Television; HP Laptop; Canon Camera; Apple iPhone. **Formula / rule first.** Manufactures(x , Camera) matches Manufactures(a , Camera) with $x = a$. Match Manufactures(Canon, Camera). Therefore $x = \text{Canon}$.

The answer-extraction method here is exactly the attachment's fact representation, query matching and unification procedure. Report the variable value, not merely "true".

7 Pigeonhole principle: objects, boxes and interpretation

Source: *Unit_3_Combinatorics.pdf*, sections 1–1.2.

The principle proves that repetition must occur. It need not identify which objects repeat.

$$N > k \Rightarrow \text{some box has at least two objects},$$

$$\text{some box has at least } \left\lceil \frac{N}{k} \right\rceil \text{ objects}.$$

Here N is the number of objects, $k \geq 1$ the number of categories, and $\lceil t \rceil$ is the least integer at least t . The source's steps are: identify objects; identify boxes; count each; apply the formula; interpret the shared category.

50 bicycles, 7 colours

Formula / rule first. Guaranteed occupancy $= \lceil N/k \rceil$. Objects: 50 bicycles. Boxes: 7 colours. Since $50 = 7 \cdot 7 + 1$, $\lceil 50/7 \rceil = 8$. Hence **at least eight bicycles share a colour**.

500 students and birthdays

Formula / rule first. $N > k$ forces two objects in the same box. Objects: 500 students. Boxes: birthday dates, with at most 366 possibilities. Since $500 > 366$, **two students share a birthday date**.

800,000 pine trees and their needles

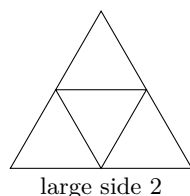
Formula / rule first. $N > k$ forces a repeated category. Objects: trees. Boxes: needle counts. If zero is allowed, counts $0, \dots, 600000$ give 600001 boxes; if all counts are positive, there are 600000. In either case $800000 > k$, so **two trees have equal needle counts**.

50 nonempty baskets, each with at most 24 apples

Formula / rule first. Guaranteed occupancy $= \lceil N/k \rceil$. Objects: 50 baskets. Boxes: the 24 counts $1, \dots, 24$. Thus $\lceil 50/24 \rceil = 3$: **three baskets have the same count**. If empty baskets are allowed, there are 25 counts and the guarantee is only $\lceil 50/25 \rceil = 2$. The nonempty condition is necessary.

Five points in an equilateral triangle of side 2

Formula / rule first. $5 > 4$ forces two points into one of four regions. Any two points in an equilateral triangle of side 1 are at distance at most 1. Join the side midpoints: four equilateral triangles of side 1 result. Assign a point on a dividing edge consistently to one region. Five points force a pair in the same small triangle. Hence their distance is at most 1.



4 small triangles, each side 1

Give the objects and boxes before the ceiling calculation. Sharing a colour, a count or a date must be translated back into the wording of the question.

8 Pigeonhole tutorials: remainders, digits and area

Source: Unit 3, Example 4 and section 1.2 Tutorials 2–5.

Among $n + 1$ integers, two differ by a multiple of n

Formula / rule first. For positive integer n , every integer is $nq + r$, $0 \leq r < n$. Equal remainders give $(nq + r) - (np + r) = n(q - p)$. There are $n + 1$ objects and n remainder boxes. Two numbers a, b have the same remainder. Write $a = nq + r$, $b = np + r$; then $a - b = n(q - p)$, so $n \mid (a - b)$. For the source's four-integer example, use $n = 3$: four integers and three boxes.

A number containing only 5s and 0s, divisible by n

Question. Prove existence for each positive integer n . **Formula / rule first.** Equal remainders modulo n imply a difference divisible by n ; more objects than remainder classes force an equal pair.

1. Take the $n + 1$ numbers $5, 55, 555, \dots$, through the number with $n + 1$ digits.
2. If one is divisible by n , it is already a valid answer.
3. Otherwise, two have the same remainder because there are only n remainder classes.
4. Subtract the smaller from the larger. The result is divisible by n and has a nonempty block of 5s followed by 0s.

For example, the subtraction pattern is $55555 - 55 = 55500$. Thus a positive number of the required form exists. Here “natural number” is interpreted as $n \geq 1$, since divisibility modulo zero is not defined.

Twelve distinct two-digit numbers

Question. Show that two differ by a two-digit number with identical digits. **Formula / rule first.** Same remainder modulo 11 gives a difference divisible by 11.

1. Put the 12 numbers into the 11 residue classes modulo 11.
2. Two share a remainder. Subtract the smaller from the larger to get a positive multiple of 11.
3. Distinct two-digit numbers lie from 10 through 99, so their positive difference is at most 89.
4. Its possible values are therefore 11, 22, 33, 44, 55, 66, 77, 88, each with identical digits.

Thirteen unit squares inside a circle of radius 2

Formula / rule first. Area of a square of side s is s^2 ; area of a circle of radius r is πr^2 . Disjoint contained figures cannot have a total area greater than their container. The 13 squares have total area $13 \cdot 1^2 = 13$. The circle has area $\pi(2)^2 = 4\pi \approx 12.566$. If all squares were pairwise disjoint, their occupied area would be 13, exceeding the circle's area. This is impossible. Hence **at least two squares have a common point**.

Practice: guarantee five cards of one suit

Formula / rule first. Maximum draws avoiding five in any suit $= 4(5 - 1) = 16$; one more forces five. There are four suits. At most four cards per suit can be drawn while avoiding five of a suit. Sixteen cards can meet that restriction, but the next cannot: 17 cards are necessary and sufficient.

The area argument follows the attachment's disjoint-area contradiction. The number-theoretic arguments use remainder boxes and subtraction.

9 Recurrences: definitions, iteration and population growth

Source: Unit 3, sections 2–3 and 6.

A **recurrence relation** (also called a difference equation in the notes) expresses a sequence term using previous terms. **Initial conditions** specify the starting values.

$$a_n = F(n, a_{n-1}, \dots, a_{n-k}).$$

Term	Meaning
Order	Largest backward index difference needed, e.g. $a_n = 3a_{n-1} - 2a_{n-2}$ has order 2.
Linear	Sequence terms occur to the first power and are not multiplied together.
Homogeneous	No separate forcing term.
Non-homogeneous	A separate forcing term occurs, e.g. $a_n = 3a_{n-1} + 2n$.
Constant coefficients	The multipliers of the sequence terms do not depend on n .

Example 6: $a_n = a_{n-1} + 3$, $a_1 = 2$

Formula / rule first. Iteration: $a_n = a_{n-j} + 3j$; choose $j = n - 1$ to reach a_1 .

$$\begin{aligned} a_n &= a_{n-1} + 3 \\ &= (a_{n-2} + 3) + 3 = a_{n-2} + 6 \\ &= a_{n-3} + 9 = \dots = a_1 + 3(n-1). \end{aligned}$$

Substitute $a_1 = 2$: $a_n = 3n - 1$ ($n \geq 1$). The sequence is 2, 5, 8, 11, ... The number of additions is $n - 1$, not n .

Example 7: $b_n = 2b_{n-1} + 1$, $b_1 = 7$

Formula / rule first. After j substitutions, $b_n = 2^j b_{n-j} + \sum_{t=0}^{j-1} 2^t$; the geometric sum is $\sum_{t=0}^{m-1} r^t = (1-r^m)/(1-r)$ for $r \neq 1$.

$$\begin{aligned} b_n &= 2b_{n-1} + 1 = 2(2b_{n-2} + 1) + 1 \\ &= 2^2 b_{n-2} + 2 + 1 \\ &= \dots = 2^{n-1} b_1 + (1 + 2 + \dots + 2^{n-2}) \\ &= 7 \cdot 2^{n-1} + (2^{n-1} - 1). \end{aligned}$$

Hence $b_n = 8 \cdot 2^{n-1} - 1$ ($n \geq 1$). At $n = 1$ the geometric tail is empty and the formula gives 7.

Application: bacteria doubling every hour

Question. A culture starts with five bacteria. Model and solve. **Formula / rule first.** Doubling gives $B_n = 2B_{n-1}$ with $B_0 = 5$; iteration gives $B_n = 2^n B_0$. Let B_n be the population after n hours. Substitute repeatedly:

$$B_n = 2B_{n-1} = 2^2 B_{n-2} = \dots = 2^n B_0 = \boxed{5 \cdot 2^n}.$$

The first populations are 5, 10, 20, 40, ...

Preserve the starting index in the source. Starting from a_1 requires $n - 1$ substitutions; starting from a_0 requires n .

10 Characteristic equations: distinct roots

Source: Unit 3, section 4.1, Example 13 and classwork CW 1.

Method. Assume $a_n = r^n$, substitute, solve the characteristic equation, write the general solution, then use the given initial indices to determine constants.

$$a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k} \implies r^k - c_1 r^{k-1} - \cdots - c_k = 0.$$

For distinct roots $r_1, \dots, r_k$, $a_n = A_1 r_1^n + \cdots + A_k r_k^n$. Each A_i is a constant fitted to the initial data.

Example 8: $C_n = 3C_{n-1} - 2C_{n-2}$; $C_1 = 5, C_2 = 3$

Formula / rule first. Distinct roots give $C_n = Ar_1^n + Br_2^n$. Assume $C_n = r^n$. Then $r^2 = 3r - 2$, hence $(r - 1)(r - 2) = 0$. Thus $C_n = A + B2^n$.

$$A + 2B = 5, \quad A + 4B = 3 \implies 2B = -2, \quad B = -1, \quad A = 7.$$

Therefore $C_n = 7 - 2^n$ for $n \geq 1$. Check: $C_1 = 5, C_2 = 3$.

Example 9: $a_n = 4a_{n-1} + 5a_{n-2}$; $a_1 = 2, a_2 = 6$

Formula / rule first. Assume $a_n = r^n$ and use the distinct-root solution. The equation is $r^2 - 4r - 5 = (r - 5)(r + 1) = 0$. Hence $a_n = A5^n + B(-1)^n$.

$$5A - B = 2, \quad 25A + B = 6 \implies 30A = 8, \quad A = \frac{4}{15}.$$

From $5A - B = 2$, $B = \frac{4}{3} - 2 = -\frac{2}{3}$. Therefore

$$a_n = \frac{4}{15}5^n - \frac{2}{3}(-1)^n.$$

The negative root produces an alternating contribution; retain $(-1)^n$ with parentheses.

Example 13: $a_{n+2} + 2a_{n+1} - 3a_n = 0$; $a_0 = 1, a_1 = 2$

Formula / rule first. With $a_n = r^n$, divide by r^n to get $r^2 + 2r - 3 = 0$; use distinct-root terms. Factor $(r - 1)(r + 3) = 0$, so $a_n = A + B(-3)^n$. Initial conditions give

$$A + B = 1, \quad A - 3B = 2 \implies -4B = 1, \quad B = -\frac{1}{4}, \quad A = \frac{5}{4}.$$

Thus $a_n = \frac{5}{4} - \frac{1}{4}(-3)^n.$

CW 1: $a_n - 7a_{n-1} + 10a_{n-2} = 0$; $a_0 = 0, a_1 = 3$

Formula / rule first. Characteristic equation $r^2 - 7r + 10 = 0$; distinct roots give a sum of exponential terms. $(r - 5)(r - 2) = 0$, so $a_n = A5^n + B2^n$. From $A + B = 0$ and $5A + 2B = 3$, use $B = -A$ to get $3A = 3$. Hence $A = 1, B = -1$ and $a_n = 5^n - 2^n$.

The attachment sometimes uses r as a sequence index. Here n is consistently the index and r the trial root, preventing two meanings in the same calculation.

11 Repeated roots, Fibonacci and direct verification

Source: Unit 3, sections 4.2–4.3 and CW 2, 4–5.

Formula / rule first. A root r of multiplicity m contributes $(A_0 + A_1n + \cdots + A_{m-1}n^{m-1})r^n$. For a double root this is $(A + Bn)r^n$.

Examples 10–12 and CW 2: fit the repeated-root constants

Example 10. $a_n = 2a_{n-1} - a_{n-2}$, $a_1 = 1.5, a_2 = 3$. **Formula / rule first.** Double root r gives $(A + Bn)r^n$. The polynomial is $(r - 1)^2$, so $a_n = A + Bn$. From $A + B = 1.5$, $A + 2B = 3$, obtain $B = 1.5, A = 0$. Thus $a_n = 1.5n$.

Example 11. $a_n = 6a_{n-1} - 9a_{n-2}$, $a_0 = 1, a_1 = 6$. **Formula / rule first.** Use $(A + Bn)r^n$ for the double root. $(r - 3)^2 = 0$ gives $a_n = (A + Bn)3^n$. The initials give $A = 1$, $3(1 + B) = 6$, hence $B = 1$: $a_n = (n + 1)3^n$.

Example 12. $d_n = 4(d_{n-1} - d_{n-2})$, $d_0 = d_1 = 1$. **Formula / rule first.** Double-root solution $(A + Bn)r^n$. The polynomial is $(r - 2)^2$. Thus $d_n = (A + Bn)2^n$, $A = 1$ and $2(1 + B) = 1$, so $d_n = (1 - n/2)2^n$.

CW 2. Same polynomial $(r - 2)^2$, with $a_0 = 1, a_1 = 6$. **Formula / rule first.** $a_n = (A + Bn)2^n$. Initials give $A = 1$, $2(1 + B) = 6$, so $B = 2$ and $a_n = (1 + 2n)2^n$.

Example 14: Fibonacci recurrence

Question. Solve $F_n = F_{n-1} + F_{n-2}$, $F_0 = 0, F_1 = 1$. **Formula / rule first.** Characteristic equation $r^2 - r - 1 = 0$; roots are $r = (1 \pm \sqrt{5})/2$. Distinct roots give $F_n = Ar_1^n + Br_2^n$. Let $r_1 = (1 + \sqrt{5})/2$, $r_2 = (1 - \sqrt{5})/2$. At $n = 0$, $A + B = 0$, so $B = -A$. At $n = 1$, $A(r_1 - r_2) = 1$. Since $r_1 - r_2 = \sqrt{5}$, $A = 1/\sqrt{5}$, $B = -1/\sqrt{5}$:

$$F_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}.$$

CW 4–5: verify proposed sequences by substitution

Question. Does $a_n = 3n$ satisfy $a_n = 2a_{n-1} - a_{n-2}$? **Formula / rule first.** Substitute the proposed formula into both sides using $a_{n-1} = 3(n - 1)$ and $a_{n-2} = 3(n - 2)$.

$$2[3(n - 1)] - 3(n - 2) = 6n - 6 - 3n + 6 = 3n = a_n.$$

Yes. **For $a_n = 5$:** the same substitution rule gives $a_{n-1} = a_{n-2} = 5$; the right side is $2(5) - 5 = 5 = a_n$. Thus the constant sequence also satisfies the recurrence.

Repeated roots require the factor n . To verify a proposed sequence, substitute its shifted terms directly; there is no need to solve the recurrence again.

12 Non-homogeneous recurrences: homogeneous plus particular

Source: Unit 3, section 5 and Examples 15–16.

The homogeneous solution a_n^h solves the recurrence with forcing removed. A particular solution a_n^p satisfies the full equation. The attachment's notation is

$$a_n = a_n^h + a_n^p.$$

Procedure. Move sequence terms to one side; remove the forcing and solve the homogeneous part; choose a particular trial; substitute into the full recurrence; match coefficients; add the two solutions; use initial values last.

Forcing $F(n)$	Trial without overlap	Adjustment for overlap
Constant c	A	Multiply by n^m if 1 is a root of multiplicity m .
Linear polynomial	$An + B$	Multiply the whole trial by n^m if 1 is such a root.
Quadratic polynomial	$An^2 + Bn + C$	Same rule with base 1.
s^n	As^n	Use $n^m As^n$ if s is a root of multiplicity m .
$P_t(n)s^n$	$Q_t(n)s^n$	Use $n^m Q_t(n)s^n$ for a matching root s .

P_t, Q_t are degree- t polynomials; all coefficients of Q_t are to be determined. A polynomial forcing alone has exponential base 1.

Example 15: $a_n = 3a_{n-1} + 2n, a_1 = 3$

Formula / rule first. $a_n = a_n^h + a_n^p$; a linear forcing with no overlap uses $a_n^p = An + B$.

1. Homogeneous root: $r = 3$, so $a_n^h = C3^n$.
2. Since 1 is not a root, substitute $a_n^p = An + B$:

$$An + B = 3[A(n-1) + B] + 2n = (3A+2)n - 3A + 3B.$$

3. Compare coefficients: $A = 3A + 2 \Rightarrow A = -1$. Compare constants: $B = -3A + 3B \Rightarrow B = -3/2$.
4. Add: $a_n = C3^n - n - 3/2$.
5. Use $a_1 = 3$: $3C - 1 - 3/2 = 3$, so $C = 11/6$.

$$a_n = \frac{11}{6}3^n - n - \frac{3}{2} \quad (n \geq 1).$$

Example 16: $a_n = 5a_{n-1} - 6a_{n-2} + 7n$

Question. Find the general solution; no initial conditions are given. **Formula / rule first.** $a_n = a_n^h + a_n^p$; use $a_n^p = Cn + D$ because base 1 does not match the roots. The homogeneous equation has $(r-2)(r-3) = 0$, so $a_n^h = A2^n + B3^n$. Substitution of the particular trial gives

$$(Cn + D) - 5[C(n-1) + D] + 6[C(n-2) + D] = 2Cn + (2D - 7C) = 7n.$$

Thus $2C = 7$, so $C = 7/2$; $2D - 7C = 0$, so $D = 49/4$. Therefore

$$a_n = A2^n + B3^n + \frac{7}{2}n + \frac{49}{4}.$$

A, B remain arbitrary because no initial values were supplied.

13 Particular solutions: overlap and repeated-root forcing

Source: Unit 3, Examples 17–18 and CW 3.

Example 17: $a_n - 2a_{n-1} = 3 \cdot 2^n$

Question. Find a particular solution. **Formula / rule first.** If forcing base s is a characteristic root of multiplicity m , multiply the ordinary trial by n^m . The homogeneous root is 2, with multiplicity 1. Thus use $a_n^p = An2^n$, not $A2^n$. Its preceding value is $A(n-1)2^{n-1}$, so

$$An2^n - 2A(n-1)2^{n-1} = A2^n[n - (n-1)] = A2^n.$$

Equating to $3 \cdot 2^n$ gives $A = 3$, hence $\boxed{a_n^p = 3n2^n}$.

Example 18: $a_n - 4a_{n-1} + 4a_{n-2} = (n+1)2^n$

Question. Find a particular solution using the attachment's substitution. **Formula / rule first.** The polynomial-times-exponential trial is $Q_1(n)2^n$. Since $(r-2)^2 = 0$, multiply it by n^2 :

$$a_n^p = n^2(An + B)2^n = (An^3 + Bn^2)2^n.$$

Let $y_n = An^3 + Bn^2$. Dividing the recurrence by 2^n produces

$$y_n - 2y_{n-1} + y_{n-2} = n + 1.$$

The required differences, obtained by expanding the powers, are

$$n^3 - 2(n-1)^3 + (n-2)^3 = 6n - 6, \quad n^2 - 2(n-1)^2 + (n-2)^2 = 2.$$

Thus $A(6n-6) + 2B = n+1$. Compare coefficients: $6A = 1$, hence $A = 1/6$; $-6A + 2B = 1$, hence $B = 1$. Therefore

$$\boxed{a_n^p = n^2 \left(1 + \frac{n}{6}\right) 2^n}.$$

The question asks for a particular solution; no initial values are needed.

CW 3: $a_n + 6a_{n-1} + 9a_{n-2} = 3$, $a_0 = 0$, $a_1 = 1$

Formula / rule first. $a_n = a_n^h + a_n^p$; repeated root r gives $(A + Bn)r^n$; constant forcing permits a constant trial if 1 is not a root.

1. $(r+3)^2 = 0$, so $a_n^h = (A + Bn)(-3)^n$.
2. Try $a_n^p = C$. Then $C + 6C + 9C = 3$, hence $C = 3/16$.
3. Add: $a_n = (A + Bn)(-3)^n + 3/16$.
4. At $n = 0$: $A + 3/16 = 0$, so $A = -3/16$.
5. At $n = 1$: $-3(A + B) + 3/16 = 1$, so $A + B = -13/48$.
6. Since $A = -9/48$, $B = -4/48 = -1/12$.

$$\boxed{a_n = \left(-\frac{3}{16} - \frac{n}{12}\right)(-3)^n + \frac{3}{16}}.$$

Apply initial values to the sum $a_n^h + a_n^p$. In the repeated-overlap example, multiply the whole degree-one trial by n^2 , exactly as in the attachment.

14 Generating functions: encode a sequence and use geometric series

Source: Unit 3, sections 7–7.2 and Examples 19–23.

An ordinary generating function records a sequence as coefficients:

$$G(x) = a_0 + a_1x + a_2x^2 + \cdots = \sum_{n=0}^{\infty} a_nx^n.$$

The coefficient of x^n is a_n ; x is the series variable and n the term index.

Sequence, starting at a_0	Ordinary generating function
$1, 1, 1, \dots$	$1/(1-x)$
$1, a, a^2, a^3, \dots$	$1/(1-ax)$
$1, -1, 1, -1, \dots$	$1/(1+x)$
$0, 1, 2, 3, \dots$	$x/(1-x)^2$
$1, 2, 3, 4, \dots$	$1/(1-x)^2$
$1, 0, -1, 0, 1, 0, -1, \dots$	$1/(1+x^2)$

Example 19: read coefficients

Question. What sequence is generated by $3+0x+8x^2+\frac{1}{100}x^3+\cdots$? **Formula / rule first.** In $G(x) = \sum_{n \geq 0} a_nx^n$, take the coefficient of each power in index order. $a_0 = 3$, $a_1 = 0$, $a_2 = 8$, $a_3 = 1/100$. Thus the sequence begins $\boxed{3, 0, 8, 1/100, \dots}$. The zero coefficient retains its position.

Example 20: $1, a, a^2, \dots$

Formula / rule first. Geometric-series identity: $1 + y + y^2 + \cdots = 1/(1-y)$. Write $G(x) = 1 + ax + a^2x^2 + \cdots$. Take $y = ax$, so $\boxed{G(x) = 1/(1-ax)}$. Here a is a fixed number, not the varying sequence term a_n .

Example 21: six ones

Formula / rule first. $1 + x + \cdots + x^{m-1} = (1-x^m)/(1-x)$. The generating polynomial is $G(x) = 1 + x + x^2 + x^3 + x^4 + x^5$. There are six terms, so $m = 6$ and $\boxed{G(x) = (1-x^6)/(1-x)}$. Higher coefficients are zero.

Examples 22–23: alternating and zero-interleaved sequences

For $1, -1, 1, -1, \dots$: Formula / rule first. Use the geometric identity with ratio $-x$. $G(x) = 1 - x + x^2 - x^3 + \cdots = 1/[1 - (-x)] = \boxed{1/(1+x)}$.

For $1, 0, -1, 0, 1, 0, -1, \dots$: Formula / rule first. Use the geometric identity with ratio $-x^2$. $G(x) = 1 - x^2 + x^4 - x^6 + \cdots = \boxed{1/(1+x^2)}$. Odd powers have coefficient zero.

Source practice: for $1, \frac{1}{2}, \frac{1}{4}, \dots$, use the same formula with $a = 1/2$, giving $1/(1-x/2)$. The source also lists the exponential generating function of all ones as e^x , but provides no exponential-generating-function procedure.

15 Generating functions: operations, shifts and differencing

Source: Unit 3, sections 7.3–7.5.

Let $F(x) = \sum_{n \geq 0} a_n x^n$ and $G(x) = \sum_{n \geq 0} b_n x^n$.

- **Equality:** $F(x) = G(x)$ exactly when $a_n = b_n$ for every $n \geq 0$.
- **Addition:** $F(x) + G(x) = \sum_{n \geq 0} (a_n + b_n) x^n$.
- **Multiplication:** the coefficient of x^n in $F(x)G(x)$ is

$$a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0 = \sum_{j=0}^n a_j b_{n-j}.$$

Shift formulas from the attachment

If $G(x)$ generates $(a_0, a_1, a_2, \dots)$, then

$$\begin{aligned} xG(x) &\text{ generates } (0, a_0, a_1, \dots), \\ x^k G(x) &\text{ shifts the sequence } k \text{ places right,} \\ \frac{G(x) - a_0}{x} &\text{ generates } (a_1, a_2, a_3, \dots), \\ \frac{G(x) - a_0 - a_1 x - \cdots - a_{k-1} x^{k-1}}{x^k} &\text{ generates } (a_k, a_{k+1}, \dots). \end{aligned}$$

Remove the first k terms before division by x^k .

	constant	x	x^2	x^3	x^4
$G(x)$	a_0	a_1	a_2	a_3	a_4
$xG(x)$	0	a_0	a_1	a_2	a_3
$(1-x)G(x)$	a_0	$a_1 - a_0$	$a_2 - a_1$	$a_3 - a_2$	$a_4 - a_3$

This makes the attachment's **differencing method** visible: multiplication by $(1-x)$ leaves the first term and then successive differences.

Example 24: generating function of $1, 3, 5, 7, 9, \dots$

Formula / rule first. Subtract $xA(x)$ from $A(x)$, then use $1 + x + x^2 + \cdots = 1/(1-x)$.

$$\begin{aligned} A(x) &= 1 + 3x + 5x^2 + 7x^3 + \cdots, \\ xA(x) &= x + 3x^2 + 5x^3 + \cdots, \\ (1-x)A(x) &= 1 + 2x + 2x^2 + 2x^3 + \cdots \\ &= 1 + 2x(1 + x + x^2 + \cdots) \\ &= 1 + \frac{2x}{1-x} = \frac{1+x}{1-x}. \end{aligned}$$

Divide by $(1-x)$:

$$A(x) = \frac{1+x}{(1-x)^2}.$$

This is the source's subtraction/differencing derivation. Keep the aligned powers and the constant term when subtracting the series.

16 Solving recurrences with generating functions

Source: Unit 3, section 8, Examples 25–27.

Example 25: $a_n = 3a_{n-1} + 2$, $a_0 = 1$

Formula / rule first. Let $A(x) = \sum_{n=0}^{\infty} a_n x^n$. For sums starting at $n = 1$,

$$\sum_{n \geq 1} a_n x^n = A - 1, \quad \sum_{n \geq 1} a_{n-1} x^n = xA, \quad \sum_{n \geq 1} x^n = \frac{x}{1-x}.$$

Multiply the recurrence by x^n and sum for $n \geq 1$:

$$A - 1 = 3xA + \frac{2x}{1-x} \Rightarrow A(1-3x) = \frac{1+x}{1-x} \Rightarrow A = \frac{1+x}{(1-x)(1-3x)}.$$

Use partial fractions as in the source:

$$A = -\frac{1}{1-x} + \frac{2}{1-3x}.$$

The identity $(1-rx)^{-1} = \sum_{n \geq 0} r^n x^n$ gives $\boxed{a_n = -1 + 2 \cdot 3^n}$. The initial value is $-1 + 2 = 1$.

Example 26: $a_n = a_{n-1} + 2$, $a_1 = 3$

Formula / rule first. Start the series at the given index: $A(x) = \sum_{n=1}^{\infty} a_n x^n$. Then

$$\sum_{n \geq 2} a_n x^n = A - 3x, \quad \sum_{n \geq 2} a_{n-1} x^n = xA, \quad 2 \sum_{n \geq 2} x^n = \frac{2x^2}{1-x}.$$

Multiply by x^n and sum for $n \geq 2$:

$$A - 3x = xA + \frac{2x^2}{1-x} \Rightarrow A = \frac{3x}{1-x} + \frac{2x^2}{(1-x)^2}.$$

Since $(1-x)^{-1}$ has coefficients 1 and $(1-x)^{-2}$ has coefficients 1, 2, 3, ..., the coefficient of x^n , $n \geq 1$, is $3 + 2(n-1)$. Thus $\boxed{a_n = 2n + 1}$.

Example 27: preserve the solution method printed in the notes

Question. Solve $a_{n+2} - 5a_{n+1} + 6a_n = 2$, $a_0 = 1, a_1 = -1$. Although placed under generating-function problems, the attachment solves this by homogeneous plus particular solutions; that is the method retained here.

Formula / rule first. $a_n = a_n^h + a_n^p$; distinct roots give exponential terms; constant forcing permits a constant trial when 1 is not a root. The characteristic polynomial is $(r-2)(r-3)$, so $a_n^h = A2^n + B3^n$. Try $a_n^p = C$: $C - 5C + 6C = 2C = 2$, hence $C = 1$. Therefore

$$a_n = A2^n + B3^n + 1, \quad A + B = 0, \quad 2A + 3B = -2.$$

With $B = -A$, $-A = -2$, so $A = 2, B = -2$. Thus $\boxed{a_n = 2 \cdot 2^n - 2 \cdot 3^n + 1}$.

Index corrections only. In Example 25, the sums must begin at $n = 1$. In Example 26, the generating function begins at $n = 1$, and the forcing sum is $2 \sum_{n \geq 2} x^n$. The printed attachment has inconsistent indices/terms in these lines; the series equations and final answers above retain its intended method.